MAIN: ASS 1

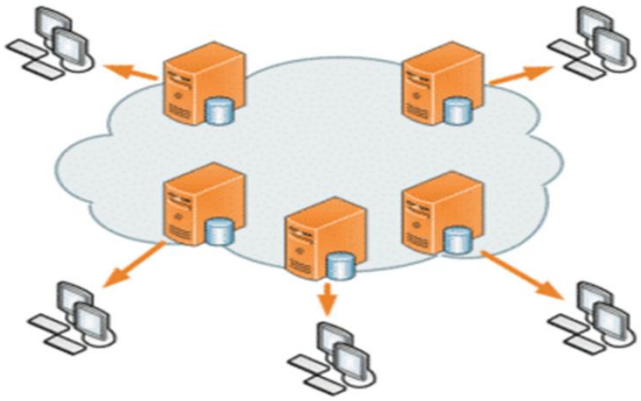
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CODE SUBJECT:	**CMPn421**
SUBJECT:	**Distributed Computer Systems**
SPECIALIZATION:	Computer engineering

DCS SHEET#1

1. Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?

A:

Define:



- A DCS is one in which **components** located at networked computers communicate and **coordinate** their actions only by-passing **messages**.
- It requires autonomous computers to be interconnected through a network.
- Each computer has to be equipped with DOS software, which enables the computers to coordinate activities and to share resources in a controlled way
- DCS requires transparency of distribution for computer users. Users shall not be aware of the fact that the system is distributed

Pros over centralized system:

- **Scalability:**
  One of the primary advantages of DCS is scalability.
  As data volumes grow, organizations can expand their processing capabilities by adding more nodes to the network.
  This scalability ensures that the system can handle increasing workloads without a significant drop in performance, providing a flexible and adaptive solution to the challenges posed by big data.
- **Fault Tolerance:**
  DCSs inherently offer improved fault tolerance compared to centralized systems.
  In a distributed environment, if one node fails, the remaining nodes can continue processing data, reducing the risk of a complete system failure.
  This resilience is crucial for maintaining uninterrupted data operations in mission-critical applications.
- **Performance:**
  Parallel processing, a key feature of DCS, contributes to enhanced performance.
  By breaking down complex tasks into smaller subtasks distributed across nets, the system can process data more quickly and efficiently.
  This results in reduced processing times and improved overall performance, enabling organizations to derive insights from data in a timely manner
- **Efficient Handling of Large Volumes of Data:**
  In the era of big data, efficiently handling large volumes of data is a paramount concern for organizations.
  DCS cope with this aspect by employing data partitioning strategies.
  Large data sets are divided into smaller, more manageable segments, and each segment is processed independently across distributed nodes.
  This approach not only accelerates data processing through parallelism and clustering but also optimizes use of resources. Each node focuses on a specific subset of the data, ensuring that the system operates efficiently and effectively.
  The ability to efficiently handle large volumes of data positions organizations to extract meaningful insights, identify patterns, and make informed decisions.

2. Define data centers and mention its applications?

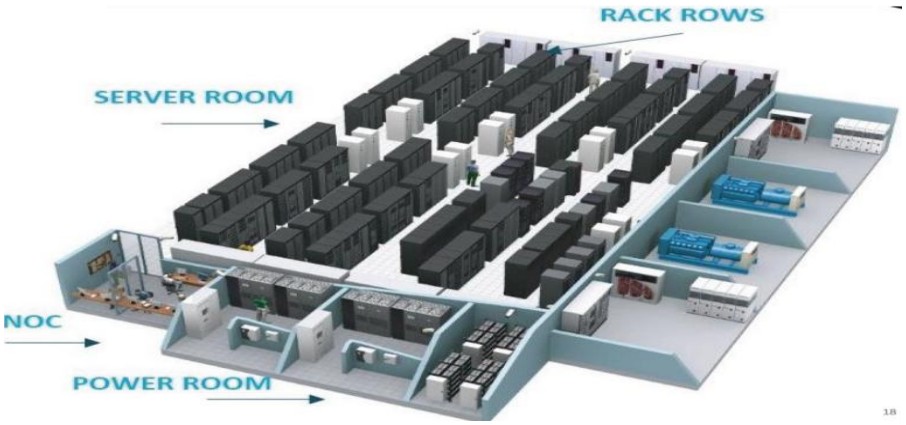
A:

- Data Centers DCs are the physical facilities that house the IT equipment and infrastructure necessary to store, process, and distribute vast and ever-growing volume of data.
- The housed and supported IT equipment including: servers, storage systems, network devices, cybersecurity devices, and other infrastructure components.
- DCs provide the physical infrastructure and environmental conditions necessary for the secured, reliable and efficient operation of these IT systems.
- DCs are typically designed with multiple levels of redundancy to ensure that they can continue to operate even in the event of a power outage, hardware failure, or other disruption.
- They employ advanced security physical and logical measures to protect the data and equipment from unauthorized access and cyber threats

3. Mention Data Centers components?

A:

- Human Resources
- Physical Suitable Site
- Racks and enclosures
- Power Distribution Units PDUs
- Cooling systems
- Fire suppression systems
- Security systems
- Computer resources
- Storage systems
- Network equipment
- Software and virtualization
- Data Center Management



4. Explain the key components of distributed computing systems using drawing?

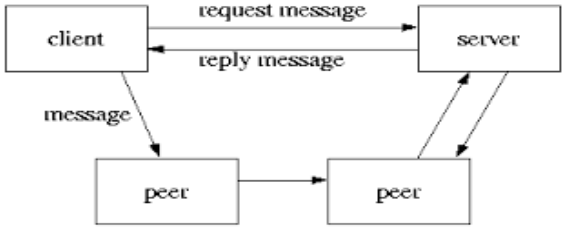
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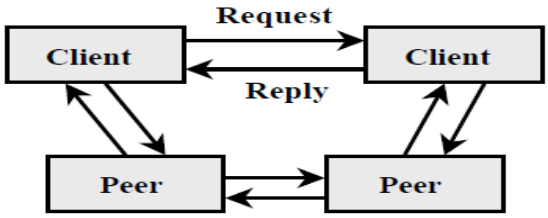


- **Geographical Distribution**:
  - Nodes are spread across different physical locations.
  - They communicate via a network, such as a local area network (LAN) or the internet.
- **Resource Sharing**:
  - Nodes share resources such as processing power, storage, and data.
  - This enables more efficient utilization of resources.
- **Concurrency**:
  - Multiple nodes operate concurrently, performing tasks simultaneously.
  - This parallelism enhances the system's overall performance and throughput.
- **Scalability**:
  - Easy to scale by adding more nodes to the system.
  - System capacity and performance improve with the addition of resources.
- **Fault Tolerance** :
  - Designed to handle failures gracefully.
  - Redundancy and replication ensure the system remains operational even if some nodes fail.
- **Transparency**:
  - The complexity of the distributed system is hidden from users.
  - Users interact with the system as if it were a single entity.

5. With the help of a diagram explain the working of distributed systems.

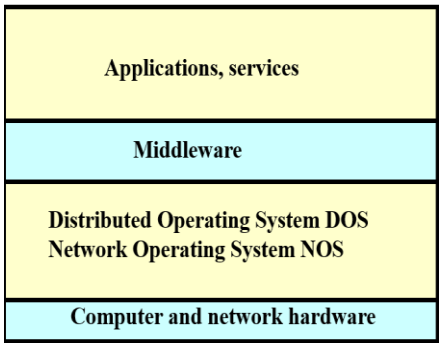
A:





6. Draw the DCS software architecture and compare between distributed operating system and network operating system?

A:



A NOS is an operating system that is expected to support networking functionalities.

This enables to connect several computers and to share many resources, e.g. files, applications, and printers.

Although each computer works self-contained it can communicate with other users in the network through the NOS.

NOS Examples: MS SERVER, IBM AIX, ORCLE SOLARES, LINUX SERVER, UNIX

- Advantages of Network OS:
- Resource Sharing
- Centralized Control
- Scalability
- NOS Security

DOS is responsible for coordinating the computers/nodes that are connected and operates in cooperation like a single large machine.

In DOS every computer operates as an individual OS, a DOS unifies all computers and they work together.

DOS EXAMPLES: MS AZURE, GOOGLE FILE SYSTEM GFS, LOCUS, ORCLE SOLARES

Advantages of Distributed OS:

- Resource Utilization
- Transparency
- Fault Tolerance
- Scalability
- DOS Security

7. Compare between centralized, decentralized, and distributed computing systems?

A:

ASPECT	CENTRALIZED SYSTEMS	DECENTRALIZED SYSTEMS	DISTRIBUTED SYSTEMS
DEFINITION	Single central server controls and manages all operations	Multiple nodes with independent control, no central authority.	Multiple interconnected nodes working together as a single system.
CONTROL	Centralized control with a single point of management	Distributed control, each node operates independently.	Shared control, nodes collaborate to achieve common goals.
SINGLE POINT OF FAILURE	High risk; if the central server fails, the whole system fails.	Reduced risk; failure of one node does not impact the entire system.	Reduced risk; designed for fault tolerance and redundancy.
SCALABILITY	Limited scalability, can become a bottleneck	More scalable, can add nodes independently.	Highly scalable, can add more nodes to distribute the load.
RESOURCE UTILIZATION	Central server resources are heavily utilized	Resources are spread across multiple nodes.	Efficient resource sharing across nodes.
PERFORMANCE	Can be high initially but may degrade with increased load	Generally good, performance improves with more nodes.	High performance due to parallel processing and resource sharing.
MANAGEMENT	Easier to manage centrally.	More complex, requires managing multiple nodes.	Complex, requires coordination and management of many nodes.
LATENCY	Lower latency, as operations are managed centrally.	Can vary, depends on the distance between nodes.	Potentially higher latency due to network communication.

8. Identify the various challenges in the distributed system.

A:

- **Heterogeneity:** different hardware, OS, networks and languages must interoperate.
- **Openness:** ability to extend/re-implement through published interfaces.
- **Security:** confidentiality, integrity and availability over open networks.
- **Scalability:** staying efficient as users/resources grow.
- **Failure handling:** detecting, masking and recovering from partial failures.
- **Concurrency:** safe simultaneous access to shared resources.
- **Transparency:** hiding distribution from users and programmers.
- **Resource sharing:** controlled sharing of hardware, data and services.
- **Synchronization / time:** no global clock; keeping data consistent.

Quality of service: performance, reliability and availability guarantees

9. Explain what is meant by (distribution) transparency, and give examples of different types of transparency.

A:

- Transparency ensures that the DCS should be perceived as a single entity by the users or the application programmers rather than a collection of autonomous systems, which is cooperating.
- The user should be unaware of where the services are located and the transfer from a local machine to a remote one should be transparent

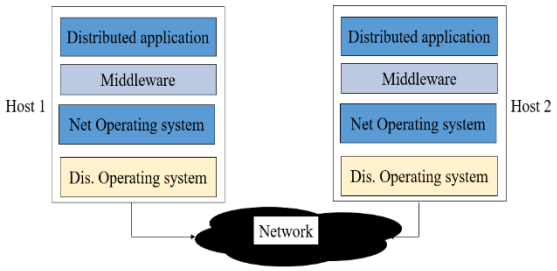
TRANSPARENCY	DESCRIPTION
Access	• Hide differences in data representation and how a resource is accessed
Location	• Hide where a resource is located, users cannot tell where hardware and software resources such as Servers, printers, files, data bases are located.
Migration	• Hide that a resource may move to another location
Relocation	• Hide that a resource may be moved to another location while in use
Replication	• Hide that a resource is replicated, OS can make additional copies of files and resources without users noticing.
Concurrency	• Hide that a resource may be shared by several competitive users, The users are not aware of the existence of other users. Need to allow multiple users to concurrently access the same resource. Lock and unlock for mutual exclusion.
Failure	• Hide the failure and recovery of a resource
Concurrency Transparency	• Automatic use of parallelism without having to program explicitly. The objectives of distributed and parallel system designers

10. Explain the differences between homogeneous and heterogenous systems?
A:

Point	Homogeneous	Heterogeneous
Nodes	Same hardware, OS and software	Different hardware, OS, languages and networks
Management	Easy and uniform	Complex; needs standards/middleware
Interoperability	Simple	Requires protocols, middleware, data conversion
Cost / flexibility	Cheaper to maintain, less flexible	More flexible, can reuse existing systems
Example	A cluster of identical servers	The Internet / WWW

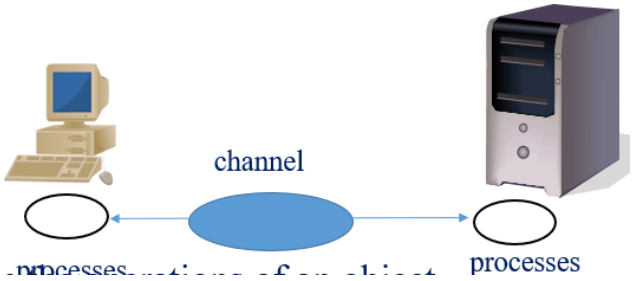
11. List the functions of middleware.
A:

- Middleware is Layer between (DOS and NOS) and distributed applications
- Hides complexity and heterogeneity of distributed system
- Bridges gap between low-level OS communications and programming language abstractions
- Provides common programming abstraction and infrastructure for distributed applications



12. Give examples for security threats and security protection.
A:
The security of a DCS has three OBJECTIVES: Confidentially, Integrity, and Availability CIA. Encryption protects shared resources and keeps sensitive information secrets when transmitted.
The security of a DS can be achieved by securing the processes and the channels used for their interactions and by protecting the objects that they encapsulate against unauthorized access
Security Threats include:

1) Threats to the processes
2) Threats to the servers
3) Threats to the communication channel
4) Denial of service DOS attacks



Examples of Security Tools:

1. **Access rights** – specify who is allowed to perform the operations of an object
2. **Principal** – associated with each session/association and each result the **authority** on which it is issued. A principal may be a user or a process

Encryption and authentication- are used to build secure channels as a service layer on top of existing communication services

13. What is meant by scalability challenge and how to achieve it?
A:

Scalability:

- One of the primary advantages of DCS is scalability.
- As data volumes grow, organizations can expand their processing capabilities by adding more nodes to the network.
- This scalability ensures that the system can handle increasing workloads without a significant drop in performance, providing a flexible and adaptive solution to the challenges posed by big data.

- **Scalability**:
    - Easier to scale by adding more nodes without overwhelming a central point.
    - Load distribution across multiple nodes improves performance and resource utilization

14. What is meant by resources sharing challenge and how to achieve it?
A:

- Resources in a distributed system are physically encapsulated within computers and can only be accessed from other computers by means of communication.
- For effective sharing, each resource must be managed by a program that offers a communication interface enabling the resource to be accessed and updated reliably and consistently.
- Users on DCS benefit from resource sharing to execute their applications.
- Hardware resources such as printers, data resources such as files, and resources are routinely share with more specific functionality such as search engines.
- users are concerned with sharing data in the form of a shared database or a set of web pages without regard for the server or servers that provide these. Patterns of resource sharing vary widely in their scope and in how closely users work together.
- A search engine on the Web provides a facility to users throughout the world, users who need never come into contact with one another directly.
- A group of users who cooperate directly share resources such as documents in a small, closed group.
- The pattern of sharing and the geographic distribution of particular users determines what mechanisms the system must supply to coordinate users‟ actions.

15. What is meant by openness challenge and how to achieve it?
A:

- The openness of the DCS is determined by the degree to which new resource-sharing services can be made available to the users.
- Open systems are characterized by the fact that their key interfaces are published.
- It is based on a uniform communication mechanism and published interface for access to shared resources.
- It can be constructed from heterogeneous hardware and software.
- Openness is concerned with extensions and improvements of DCSs.
- Detailed interfaces of components need to be published.
- New components have to be integrated with existing components.
- Differences in data representation of interface types on different processors (of different vendors) have to be resolved.

16. What is meant by fault tolerance challenge and how to achieve it?
A:

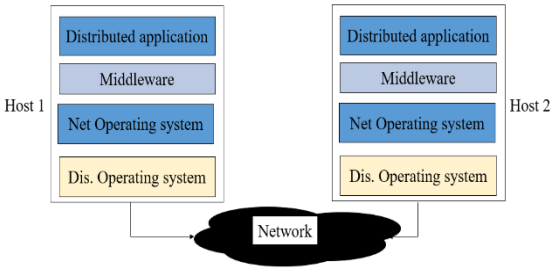
- When some faults occur in hardware and the software program, it may produce incorrect results or they may stop before they have completed the intended computation so corrective measures should to implemented to handle this case.
- Failure handling is difficult in DCS because the failure is partial i.e., some components fail while others continue to function.
- Hardware, software and networks fail!
    - Distributed systems must maintain *availability* even at low levels of hardware/software/network *reliability*.

Fault tolerance is achieved by

- recovery
- redundancy

17. Explain and List the functions of middleware.
A:

Middleware is a set of services that enables applications and end-user to interact with each other across a heterogeneous DCS by **masking the heterogeneity of the underlying networks, hardware, OS, and programming languages**



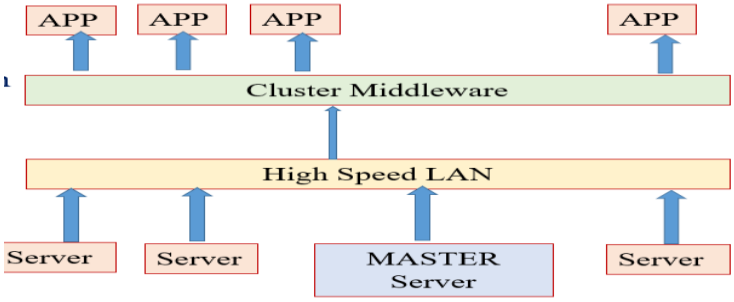
- Middleware is Layer between (DOS and NOS) and distributed applications
- Hides complexity and heterogeneity of distributed system
- Bridges gap between low-level OS communications and programming language abstractions
- Provides common programming abstraction and infrastructure for distributed applications

18. Explain and List the functions of clustering computing.
A:

- A cluster is a group of linked computers.
- Working together closely so that in many respects they form a single computer
- The components of a cluster are connected to each other through fast LANs.
- Clusters are deployed to improve performance and/or availability over that provided by a single computer, while being much more cost effective than single computers of comparable speed or availability

Cluster consists of:

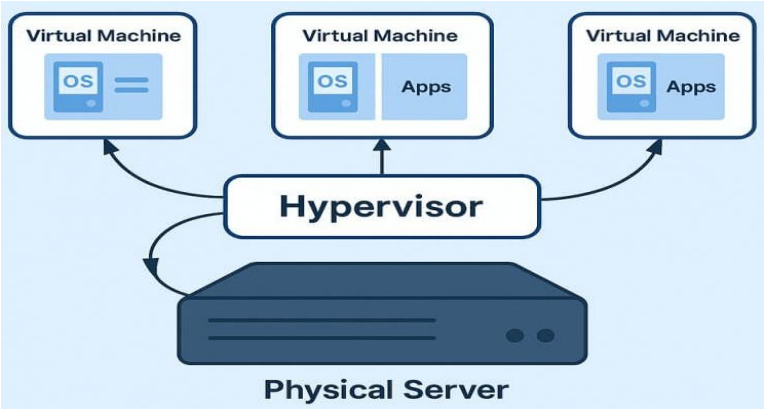
- Nodes (master + computing)
- Network
- Oss
- Cluster middleware which permits compute clustering programs to be portable to a wide variety of clusters



19. Explain and List the functions of virtualization.

A:

- Virtualization is a technique of partitioning the resources of a single computing platform into multiple segregated, virtualized, execution environments.
- Each environment runs independently of the other, thus allowing multiple OSs to run on the same hardware.
- The concept of virtualization already present in every-day computing…
- Most modern OSs contain a simplified system of virtualization;
- Each running process is able to act as if it is the only thing running.
- The CPUs and memory are virtualized.
- CPU Virtualization: If a process tries to consume all of the CPU, the OS will preempt it and allow other processes to execute; If an OS tries to consume all of the CPU, the hypervisor will preempt it and allow other processes to execute;
- Memory virtualization: a running process has its own virtual address space that the OS maps to physical memory to give the process the illusion that it is the only user of RAM
- A running OS has its own virtual address space that the hypervisor maps to physical memory to give the process the illusion that it is the only user of RAM.
- Each execution environment is called a *guest* and the computing platform on which they execute is called the *host*.



- The software enabling these multiple execution environments is commonly referred to as **Hypervisor** or Virtual Machine Monitor (VMM).
- The Hypervisor runs on the host and acts as a *bridge* between the host and the guests;
- Multiple VMs can be implemented on a single hardware platform to provide individuals or user groups with their own OS environments

20. What are the benefits of a computer cluster?

A:

- This model applies to both physical resources and more logical services:
- Physical resources such as storage and processing can be made available to networked computers, removing the need to own such resources on their own.
- Users can remotely require storage facility for file storage requirements (e.g. for multimedia data such as photographs, music or video) and/or for Desktops.
- Users can rent one or more computational nodes, either to meet their basic computing needs or indeed to perform distributed computation.
- At the other end of the spectrum, users can access sophisticated data centers.
- Software services can also be made available across the global Internet using this approach.
- Many companies now offer a comprehensive range of on-demand services for effective rental, including services such as email

21. What are the types of cluster computing?
A:

- **High availability Clusters (HA) (Linux):**
  - Mission critical applications
  - High availability /Failover clusters are implemented for the purpose of improving the availability of services which the cluster provides
  - Provides redundancy
  - Eliminate single point of failure
- **Network Load balancing Clusters:**
  - Operate by distributing a workload over multiple backend nodes
  - Typically the cluster will be configured with multiple redundant load-balancing front ends
  - All available servers process requests
  - Web servers, mail servers, ftp servers,….
- **Science Clusters:**
  - Science cluster is a computer design that uses parallel processing across multiple computers to create cheap powerful supercomputer
  - It has two types of computers, a master/front/service computer and a node computers
  - When a large problem or set of data is given, the master computer first runs a program that breaks the problem into discrete pieces, it then sends a piece to each node to compute
  - As nodes finish their tasks, the master computer sends more piece to them until the entire problem has been computed

GOOD LUCK